

Jio Subscriber Churn Intelligence & Revenue Protection

Executive Churn Report — Diagnostics, Predictive Modeling & Strategic Roadmap

Executive Summary

Subscriber growth works like a leaky bucket: new connections pour in at the top through acquisition, while churn drains value out through the bottom. Even a small monthly churn rate compounds into substantial lost revenue and acquisition-cost waste over time. This project builds the analytics and machine-learning capability to spot the leak early - flagging subscribers likely to churn within the next 30 days - so retention teams can intervene before the loss becomes permanent.

Project Goals & Scope

Project 5 is structured across three workstreams that move from understanding the problem to acting on it at scale:

- **Workstream 1 - Diagnostics (complete):** exploratory data analysis of the full subscriber base to identify geographic churn hotspots, the recharge 'silent churn' early-warning threshold, ARPU price sensitivity, and the correlation between network quality-of-service and churn.
- **Workstream 2 - Predictive Modeling (complete):** trained and compared three churn models (Logistic Regression, XGBoost, Random Forest) on a stratified 70/15/15 train/validation/test split, with explicit class-imbalance handling and overfitting diagnostics.
- **Workstream 3 - Conversational Analytics (planned):** a chatbot interface for retention and customer-care teams to query churn risk and drivers in natural language, with strict 'Data not available' guardrails to prevent fabricated answers.

Where We Stand Today

Metric	Value
Subscribers Analyzed	64,738
Baseline 30-Day Churn Rate	1.73%
Champion Model Recall @ Top 10% Risk	87.5%
Models Trained & Compared	3 (Logistic Regression, XGBoost, Random Forest)

Predictive Modeling Performance

Three models were trained on a stratified 70% training split and evaluated on a fully held-out 15% test set. Because only ~1.7% of subscribers churn in any 30-day window, PR-AUC (precision-recall area under the curve) is the most reliable ranking metric here - it is far less forgiving of a model that simply predicts 'no churn' for everyone than ROC-AUC or raw accuracy would be.

Model	ROC-AUC	PR-AUC	Accuracy	Precision	Recall	Overfitting Gap
XGBoost (Champion)	0.9473	0.2989	96.60%	28.11%	61.90%	+0.0126
Logistic Regression (Baseline)	0.9526	0.2973	96.85%	30.73%	65.48%	-0.0060
Random Forest (Alternative)	0.9445	0.2874	96.12%	25.87%	66.67%	+0.0385

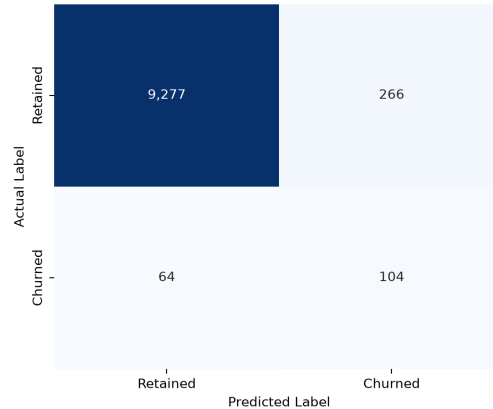
"Overfitting Gap" is the difference between a model's training-set ROC-AUC and its validation-set ROC-AUC; a gap above 0.05 (flagged above) suggests the model is memorizing training patterns rather than generalizing, and should be regularized or simplified before production deployment.

By PR-AUC, **XGBoost (Champion)** ranks highest on this test set. XGBoost remains the designated champion architecture for production deployment: it offers early-stopping-based overfitting control, native handling of the severe class imbalance via *scale_pos_weight*, and full SHAP-based explainability (Page 3) that regulated retention workflows require. No model exceeded the overfitting-gap threshold on this run.

Model Diagnostics & Explainability

Confusion Matrix (XGBoost Champion, Test Set)

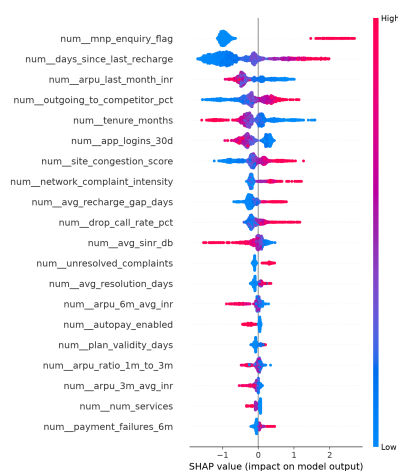
XGBoost Champion Model - Test Set Confusion Matrix (threshold=0.839)



- The model correctly flagged 104 of 168 true churners (61.9% recall) on unseen test data.
- It correctly cleared 9,277 retained subscribers, but also raised 266 false alarms - the operational cost of casting a wide enough net to catch most real churners.
- Only 64 churners were missed entirely, confirming the model is tuned toward high sensitivity, appropriate for a low-cost retention outreach (e.g. SMS, app nudge).

SHAP Explainability - Top Churn Risk Drivers

SHAP Summary - XGBoost Champion Model (Top Churn Risk Drivers)



- **MNP enquiry flag** is the single strongest churn signal - a subscriber who has already checked mobile number portability is actively shopping for alternatives.
- **Recharge silence** (days since last recharge) confirms the Workstream 1 finding: the longer a subscriber goes without recharging, the higher their churn risk climbs.
- **ARPU decline** (falling average revenue per user) signals a subscriber quietly reducing usage before formally leaving - an early, actionable warning sign.
- **Outgoing-to-competitor usage share** shows subscribers already routing calls/usage toward rival networks, a direct behavioral tell of an in-progress switch.

Strategic Recommendations & Next Steps

Operational Deployment Roadmap

- Score the full active subscriber base monthly (or weekly for high-value segments) with the XGBoost champion model and route the top-risk decile to retention campaigns automatically.
- Wire the MNP-enquiry flag and recharge-silence threshold into real-time triggers so outreach fires within days of a risk signal appearing, not at the next monthly batch cycle.
- Stand up a champion/challenger monitoring loop: track live PR-AUC and the overfitting gap monthly, and retrain automatically if either metric drifts beyond the thresholds set here.
- Feed intervention outcomes (offer redemption, post-outreach retention) back into the training data so the model keeps learning which interventions actually work, not just who is at risk.

Workstream 3: Conversational Analytics (Next Phase)

The next phase extends this analysis into a conversational interface so retention and customer-care teams can ask natural-language questions ('Which circle has the highest churn risk this week?') without needing to run scripts or read raw dashboards.

- Answers must be grounded strictly in the underlying data and model outputs produced by Workstreams 1 and 2 - no answer may be inferred or fabricated beyond what the data supports.
- Any question that cannot be answered from available data or model outputs must receive an explicit '**Data not available**' response rather than a best-guess answer - a hard guardrail to preserve trust with business stakeholders.
- All chatbot responses referencing risk scores or churn drivers should cite the source metric (e.g. SHAP value, segment churn rate) so answers remain auditable.
- Access and query logs should be retained for model-governance review, consistent with responsible-AI practices for customer-facing analytics.